

Calibration

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 | Generated: | Engine: DiscreteSim | Sections: 8

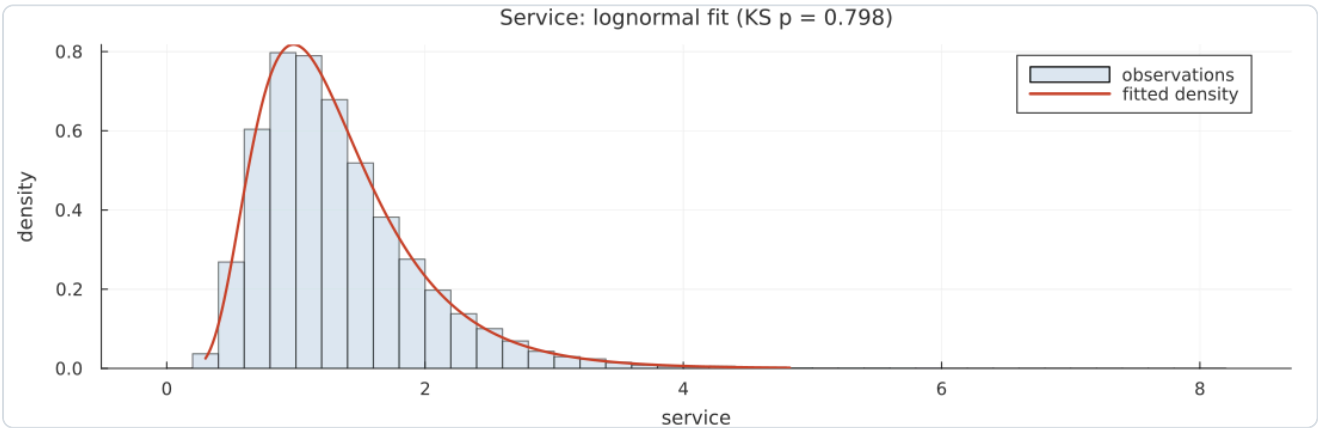
Calibration

The model parameters come from measurements, and every fit is tested against the observations it claims to describe.

Maximum-likelihood fits

Fits

Series	Family	Mean	Sample mean	Cv	N	Ks stat	Ks p	Verdict
failure_interval	weibull	172.3	172.3	0.663	354	0.024	0.988	validated
interarrival	gamma	1.197	1.197	1.005	54127	0.00293	0.741	validated
repair	lognormal	11.1	11.1	0.529	354	0.044	0.488	validated
service	lognormal	1.329	1.328	0.473	48780	0.00292	0.798	validated
demand_size	lognormal	14.3	14.4	0.488	82	0.056	0.957	validated



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What the data implies for the model

Arrival rate 0.835
Service rate 0.753
Mtbf 172.3
Mttr 11.1
Availability 0.94
Demand mean 14.3